



The effects of increased temperature on crop growth and yield of soybean grown in a temperature gradient chamber



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ABSTRACT

The global surface temperature is projected to rise and will affect crop performances. This study investigated the effects of increased temperature on yield and dry matter production in a controlled environment that mimicked field conditions using a temperature gradient chamber (TGC). Experiments were conducted from 2009 to 2012 using the soybean cultivar Enrei, which was grown in soil culture beds. Plants were grown in two TGCs as replicates for temperature treatment. Three temperature treatments, near ambient temperature (T_a), ambient temperature + 1 °C ($T_a + 1$), and ambient temperature + 3 °C, in 2009 and 2010, and ambient temperature + 2 °C, in 2011 and 2012 ($T_a + 2/T_a + 3$), were established by dividing the rows along which the temperature gradient was created. The aboveground dry matter was significantly reduced by increased temperature from 11% in 2012 to 27% in 2009. Decrease of dry matter accumulation was obvious particularly from flowering to early seed filling and it was associated with decline of leaf photosynthetic rate, stomatal conductance and the carbon isotope discrimination. Reduced pod number, reduced seed number, and, to some extent, smaller seeds led to a decreased harvest index (HI). These phenomena might be associated with the delayed pod set and lower seed growth rate under warmer treatment. Seed yield was the most responsive parameter in 2009 and 2010, and in 2011 and 2012, it was still reduced under continuously wet conditions. Combined data showed that total dry matter, seed yield, and HI were consistently reduced by increased temperature. It is suggested that the concomitant increase of vapor pressure deficit with increased temperature exacerbated the temperature effects. In addition, reduced ambient CO₂ and low light intensity as the artifacts of the facility might have accounted for the greater effect of high temperature.

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1. Introduction

Soybeans are a major oilseed crop produced and consumed throughout the world, and produce the second most important vegetable oil after rapeseed (United States Department of Agriculture, 2011). It has been recognized that there is a need to increase the yield of soybean (Board and Kahlon, 2011) along with other food crops to keep pace with the growing demand caused by growing populations and rising incomes (United Nations Environmental Programme, 2006). In addition, the global average surface

temperature will increase by between 1.4 and 5.8 °C in this century (United Nations Environmental Programme, 2006), which could potentially have negative impacts on important agronomic crops, including soybeans (Hatfield et al., 2011). Not only agronomic adaptation, such as shifting cropping seasons, but also the development of heat-tolerant genotypes through conventional breeding or the use of advanced techniques of molecular breeding and genetic engineering are promising; however, a successful strategy requires concerted efforts among different areas, including plant physiology (Wahid et al., 2007). This has led to many experiments around the globe on the effects of increased temperatures or its interactions with carbon dioxide (CO₂) on soybeans; however, considerable inconsistencies exist regarding the response of total biomass and yield to increased temperature (Tacarindua et al., 2012). Biomass production or photosynthetic activity was not significantly affected by temperatures slightly higher than 30 °C (Sato, 1976; Harley et al., 1985; Baker et al., 1989) and biomass was not considerably affected by temperatures near 40 °C (Allen and Boote, 2000). Seed yield was slightly reduced by high temperature stress induced only at the

Abbreviations: DM, dry matter; g_s , stomatal conductance; HI, harvest index; Pn, leaf photosynthetic rate; TGC, temperature gradient chamber; VPD, vapor pressure deficit.

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initial seed filling period (Ferris et al., 1999), whereas seed yield was reduced under a moderately high temperature of 27.5 °C during the entire growing season (Heinemann et al., 2006).

Temperature gradient chambers (TGCs) allow the study of temperature effects on crops under field-like conditions, where the inside temperatures tend to keep track with the ambient temperatures (Horie et al., 1995). Nevertheless, experiments conducted to evaluate the effects of temperature on soybean yield and dry matter (DM) production using TGCs are very limited and have shown varied responses to increased temperatures from moderate (Allen and Boote, 2000; Shiraiwa et al., 2006) to distinct (Ohe et al., 2007). These circumstances require further studies to better quantify the impacts as well as characterize the exact mechanisms involved in the response of biomass production and seed yield of soybeans to increased temperature. The objective of this study was therefore to investigate the effects of increased temperature on soybean growth and yield using TGCs.

2. Materials and methods

2.1. Environment description

Soybean cultivar Enrei, determinate and of maturity group IV, was grown under various temperatures in TGCs at the Experimental Farm of Kyoto University at Kyoto City, Japan (35.0° N lat., 135.5° E long., 71 m asl). The TGC, which was 2 m wide and 25 m long, created a nearly linear temperature gradient along its longitudinal axis from near ambient to a temperature that was several degrees higher, while maintaining the natural diurnal changes in air temperature (Horie et al., 1995). The chambers were covered with polyethylene terephthalate film with a light transmittance of 80%.

Soybean seedlings were transplanted on July 13 and July 21 in 2009 and 2010, respectively, and directly sown on July 12, in 2011 and 2012 into the soil culture bed in two TGCs, arranged in four rows of 0.25 m wide, 24 m long, and 0.25 m intra-row spacing. Three temperature treatments, near ambient temperature (Ta), ambient temperature + 1 °C (Ta + 1), and ambient temperature + 3 °C, in 2009 and 2010, and ambient temperature + 2 °C, in 2011 and 2012 (Ta + 2/Ta + 3), were established by dividing the rows along which the temperature gradient was created.

The TGC was equipped with an irrigation system containing a drainage pipe located 50 cm below the soil surface. Water was supplied through the pipe to raise the water table to approximately 30 cm below the soil surface, and the water was evenly distributed throughout the entire soil culture bed. The moisture content was monitored in 2010 using a moisture sensor (ECH₂O EC-5 moisture sensor; DECAGON, Pullman, Washington, USA) installed at a depth of 15 cm, whereas in 2011 and 2012, soil moisture content was monitored using a time domain reflectometry (TDR) meter (SONY Tektronix Co. Ltd., Tokyo, Japan) installed at a depth of 30 cm. The volumetric water content was maintained at approximately 22.2% and 24.2% (in 2009 and 2010, respectively) and 24% (in 2011 and 2012) throughout the entire growth period by sub-soil irrigation. Considering that the range of soil water content for plant transpiration was considered to be between 33% (field capacity) and 13% (data not shown), these values appear to be approximately one-half the level of transpirable soil water. A detailed description of the irrigation system can be found in Tacarindua et al. (2012).

The soil used is classified as alluvial sandy loam (Fluvic Endoaquepts) and the plots were kept weed free by chemical control or hand weeding.

2.2. Measurements

Dates of the developmental stages (Fehr and Caviness, 1977) were recorded and the aboveground parts of six plants per

treatment were sampled at the onset of flowering (R1), the beginning of seed filling (R5) (2011 and 2012), at 5 days after R5 at Ta (in 2009), or at 10 days after R5 at Ta (in 2010) and eight plants at harvest maturity (R8) to determine the changes in aboveground DM accumulation. At harvest maturity, plant components were separated into leaves (including petioles), stems, pod shells, and seeds that were used to calculate the final single-seed size, seed number, seed yield, and total aboveground DM.

In 2011 and 2012, photosynthetic rate (Pn) and stomatal conductance (g_s) were measured in the central leaflets of fully developed leaves of three plants per treatment (Ta and Ta + 2) once a week from near flowering, using LI-6400 (LI-COR, Inc., Lincoln, NE, USA). According to these measurements, which were conducted during the day, CO₂ concentration [CO₂] at Ta + 2 was 350 ± 13 μL/L compared to 372 ± 8 μL/L under Ta. The carbon isotope discrimination of milled samples (0.5 mg for seed and 2.0 mg for leaf) collected at harvest maturity from Ta and Ta + 2 plots was determined by mass spectrometry (Delta V; Thermo Fisher Scientific) at Kyoto University Ecological Center. Carbon isotopic composition of seed and leaf samples was expressed relative to the standard Pee Dee Formation of Belemnite.

2.3. Statistical analysis

The effects of increased air temperature on seed yield, yield components, DM accumulation, and carbon isotope discrimination (CID) were evaluated using an analysis of variance (ANOVA) with statistical software SAS version 9.3 (SAS Institute Inc., Cary, NC, USA).

3. Results

3.1. Temperature environment and vapor pressure deficit

The day, night, and mean air temperatures for the entire growing season are shown in Table 1. In the TGC, the lower temperature treatment was near the ambient temperature, and a temperature difference of approximately 2–3 °C was achieved for the day, night, and mean temperatures. Changes in the daily mean temperature during the entire growing season are shown in Fig. 1. The inside temperatures tended to track the diurnal fluctuation of the outdoor temperatures. As generally happens in soybean production in this region, the plants experienced a period of relatively high temperature from R1 to R5 in the four study years. The temperatures tended to decrease in the mid-seed filling period and then increased slightly toward physiological maturity (R7). This was followed by relatively low temperatures until harvest maturity.

Estimated day, night, and mean vapor pressure deficit (VPD) from emergence (VE) to R7 are shown in Table 2. In general, most data for daily mean VPD ranged from 0.82 to 1.37 kPa in 2009, and from 1.40 to 2.13 kPa in 2010. In 2011 and 2012, the daily mean VPD ranged from 0.9 to 1.5 kPa and the day average reached approximately 2.1 kPa before R1. The VPD tended to be higher in the warmer treatment compared to that under Ta, and if we compare the four study years, the VPD in 2010 was the highest of all years.

3.1.1. Developmental stages

Increasing air temperature by approximately 2–3 °C did not have a significant effect on R1, except in 2010, when flowering was delayed by 3 days. The number of days from R1 to R5 and the duration of seed filling (R5–R7) were significantly increased by increased temperatures (Table 1). The progress of pod setting (R1–R5) was more affected by the increase in temperature than the progress from VE to R1 and R5 to R7, particularly in 2009 and 2010. During these years, the time from R1 to R5 increased by 5

Table 1
Response in phenology and yield components of soybean cultivar Enrei to temperature.

Year and treatment	Air temperature ^a			Days from			DM at R8			Node number (/plant)	Pod number (m ⁻²)	Seed number (m ⁻²)	Seed number (/pod)	Single seed weight ^b (mg)	Yield (g m ⁻²)	HI
	Mean (°C)	Day (°C)	Night (°C)	VE-R1 (d)	R1 to R5 (d)	R5 to R7 (d)	Leaf+ petiole (g m ⁻²)	Stem + podshell + seed (g m ⁻²)	Total (g m ⁻²)							
2009																
Ta ^c	25.9	28.2	23.7	30	27	35	48.5	673.6	722	28.1	879	1626	1.8	248	403	0.56
Ta + 1	27.1	29.3	25.0	30	28	35	108.8	544.8	654	27.6	732	1318	1.8	236	305	0.47
Ta + 3	28.7	31.1	26.2	30	32	37	93.2	432.1	525	34.9	628	1060	1.7	227	241	0.46
2010																
Ta	27.1	29.5	24.7	30	14	39	38.7	578.5	617	25.1	774	1349	1.7	242	327	0.53
Ta + 1	28.3	30.7	26.0	32	14	40	74.4	488.8	563	24.3	709	1202	1.7	231	278	0.49
Ta + 3	30.1	32.6	27.6	33	19	41	147.6	412.1	560	27.7	651	977	1.5	208	203	0.36
2011																
Ta	25.7	27.9	23.4	27	19	40	57.9	805.6	864	35.1	983	1844	1.9	262	483	0.56
Ta + 1	26.6	29.0	24.2	27	22	40	138.0	742.0	880	36.4	1005	1717	1.7	242	416	0.48
Ta + 2	27.4	29.7	25.1	27	24	38	66.9	572.8	640	29.8	893	1442	1.6	233	336	0.53
2012																
Ta	26.9	29.3	24.6	25	19	37	152.3	744.9	897	37.6	910	1641	1.8	248	435	0.48
Ta + 1	27.5	30.0	25.0	25	20	38	151.9	705.3	857	35.0	895	1589	1.8	240	407	0.47
Ta + 2	28.8	31.5	26.0	25	23	39	153.8	643.9	798	33.9	818	1458	1.8	235	365	0.46
ANCOVA			F value (P)													
Temperature (T)			13.28 (<0.01)	91.57 (<0.01)	14.30 (<0.01)	5.33 (<0.05)	88.18 (<0.01)	14.29 (<0.01)	0.95 (n.s.)	29.38 (<0.01)	220.04 (<0.01)	0.56 (n.s.)	13.75 (<0.01)	171.95 (<0.01)	22.10 (<0.01)	
Year (Y)			2.5 (<0.1)	1.86 (n.s.)	10.07 (<0.01)	0.98 (n.s.)	3.86 (<0.05)	2.17 (n.s.)	3.81 (<0.05)	1.11 (n.s.)	6.54 (<0.01)	0.93 (n.s.)	0.50 (n.s.)	5.07 (<0.01)	1.49 (n.s.)	
T × Y			4.93 (<0.05)	1.42 (n.s.)	8.27 (<0.01)	0.87 (n.s.)	3.88 (<0.05)	2.06 (n.s.)	3.49 (<0.05)	1.23 (n.s.)	6.81 (<0.01)	1.03 (n.s.)	0.50 (n.s.)	5.15 (<0.05)	1.44 (n.s.)	

^a Temperature for the entire growing season.

^b Based on dry matter (DM).

^c Near ambient temperature.

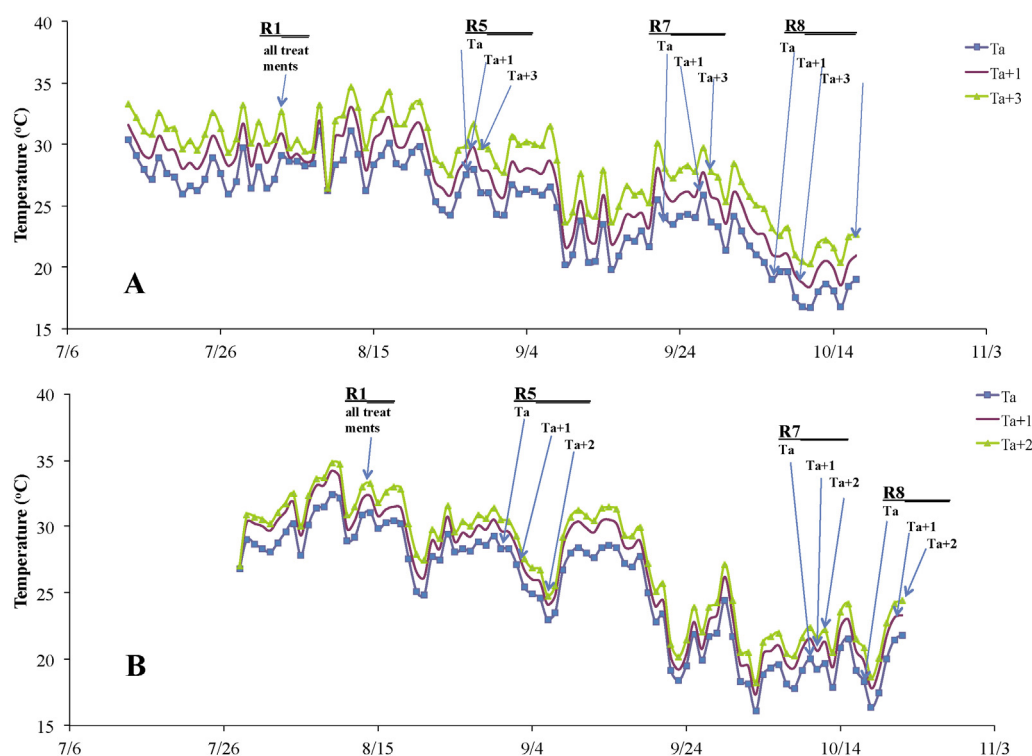


Fig. 1. Daily changes in mean air temperature at different treatments in a temperature-gradient chamber in 2009 (A) and 2011 (B).

days, whereas the time from R5 to R7 increased by 2 days. There was a significant difference in the duration of the developmental stages among different years.

3.1.2. Yield components and DM accumulation

Seed yield and yield components for all 4 years are presented in Table 1. Increased temperature had no significant effect on the number of nodes but significantly reduced the number of pods and seeds, seed size, and seed yield. The temperature increase reduced pod number by 30% and 16%, in 2009 and 2010, respectively, and by 10% in 2011 and 2012. Seed number was reduced by 35%, 28%, 22%, and 11% in 2009, 2010, 2011, and 2012, respectively. Single-seed weight reduction ranged from 5% in 2012 to 14% in 2010, with a similar trend reported by Tacarindua et al. (2012). Seed yield was

most responsive to the increase in temperature, except in 2012, when it was reduced by 16%. In 2009, seed yield was reduced by approximately 40%, while in 2010 and 2011, it was reduced by 38% and 30%, respectively.

The total DM at R8 was reduced by 27% in 2009, 9% in 2010, 26% in 2011, and 11% in 2012 (Table 1). The effect of increased temperature on DM accumulation was observed during most of the growing period and a major part of the differences occurred in the period from 35 to 65 days after sowing (DAS) in 2009, 32 to 55 DAS in 2010, 32 to 50 DAS in 2011, and 29 to 56 DAS in 2012. There were no evident differences in plant responses between Ta and Ta + 1, but there were differences between Ta and Ta + 2/Ta + 3, although they were significant only in 2010 and 2011 (Fig. 2). The ANOVA also showed that the differences in accumulated DM between Ta and

Table 2

Estimated vapor pressure deficit (VPD) during different stages in the temperature-gradient chamber.

Year	Air temperature			VPD (KPa)								
	Mean (°C)	Day (°C)	Night (°C)	VE-R1			R1-R5 + 10			R5-R7		
				Mean	Day	Night	Mean	Day	Night	Mean	Day	Night
2009												
T1	25.9	28.2	23.7	0.82	1.17	0.47	1.09	1.55	0.63	–	–	–
T2	27.1	29.3	25.0	1.11	1.44	0.79	1.27	1.69	0.85	–	–	–
T3	28.7	31.1	26.2	1.37	1.71	1.02	1.36	1.72	1.00	–	–	–
2010												
T1	27.1	29.5	24.7	1.52	2.12	0.93	1.40	1.90	0.91	–	–	–
T2	28.3	30.7	26.0	1.80	2.34	1.25	1.62	2.03	1.20	–	–	–
T3	30.1	32.6	27.6	2.13	2.68	1.58	1.95	2.30	1.59	–	–	–
2011												
T1	25.7	27.9	23.4	1.29	1.85	0.72	0.84	1.15	0.54	0.91	1.31	0.52
T2	26.6	29.0	24.2	1.41	1.98	0.85	0.90	1.18	0.62	0.96	1.35	0.57
T3	27.4	29.7	25.1	1.56	1.99	1.13	1.13	1.39	0.88	1.24	1.60	0.88
2012												
T1	26.9	29.3	24.6	1.25	1.82	0.69	0.93	1.28	0.57	0.87	1.28	0.46
T2	27.5	30.0	25.0	1.41	2.00	0.83	1.11	1.49	0.72	1.01	1.49	0.54
T3	28.8	31.5	26.0	1.54	2.16	0.92	1.24	1.65	0.83	1.16	1.65	0.67

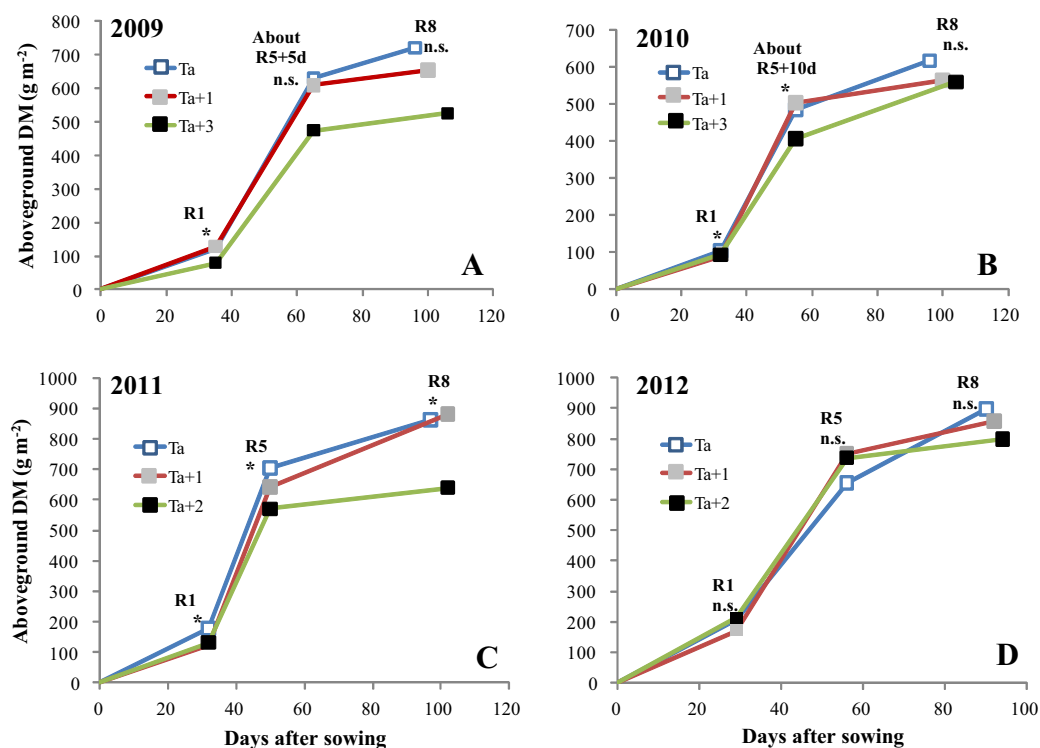


Fig. 2. Temporal changes in aboveground dry matter (DM) of soybean cv. Enrei in 2009 (A), 2010 (B), 2011 (C), and 2012 (D). The statistical significances are comparisons between Ta and Ta +2/Ta +3 of each year.

T +2/Ta +3 at R8 were significant only in 2011. Because of the effects of temperature on crop phenology, the developmental stages at the time of plant sampling differed among treatments. For example, when sampled at 65 DAS in 2009, the plants under Ta were at 5 days after R5, whereas the plants under Ta +3 were harvested at R5.

Increasing temperature also significantly affected the fraction of total aboveground DM allocated to the seed HI (Table 1). HI was reduced by approximately 18% in 2009, 32% in 2010, 6% in 2011, and 5% in 2012.

Fig. 3 shows the progress of pod development expressed by pod DM to total aboveground DM plotted against days after the R5 date under the Ta treatment in 2009 and 2010. As the conditions were warmer, pod development was late at early R5, and this was held over to late seed filling in both years, resulting in reduced HI. In Ta +3, DM of the leaf + petiole was significantly greater than under Ta, indicating leaf senescence was late in the warmer treatment (Table 1).

3.1.3. Photosynthesis and stomatal conductance

Increasing the temperature by approximately 2 °C reduced the photosynthetic rates, with 9–16% lower values under Ta +2 than under Ta (Fig. 4A). Two of the four observed reductions in photosynthetic rates appeared to be associated with the reduction of stomatal conductance (Fig. 4B).

3.1.4. Carbon isotope discrimination

The CID of seed samples ranged from 19.50 to 20.39‰ in 2011 and from 20.06 to 20.39‰ in 2012, with the values being significantly lower under warmer treatments; however, there was a significant effect of temperature on CID in leaf samples only at 10% level (Table 3).

4. Discussion

A 2–3 °C gradient was created most of the time during the growing season, and it partly affected plant phenology. The observed

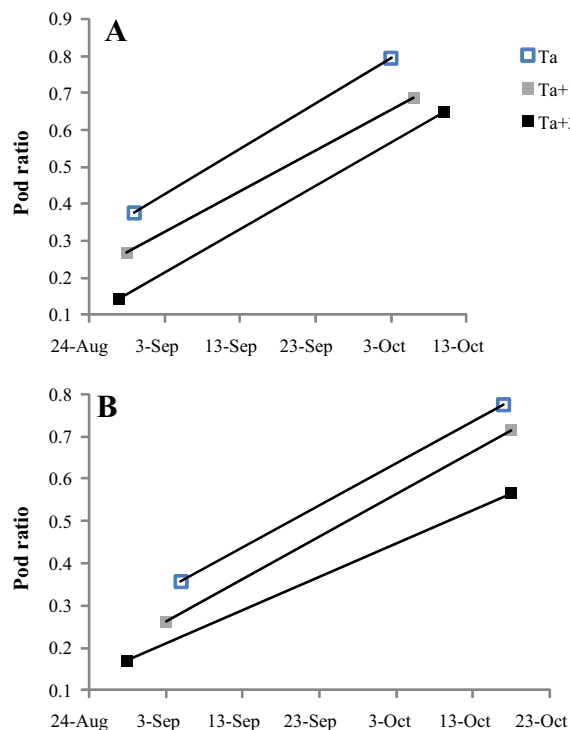


Fig. 3. Pod dry weight per total aboveground dry weight at R5 + 5 (R5 of Ta in 2009, A), R5 + 10 (R5 of Ta in 2010, B).

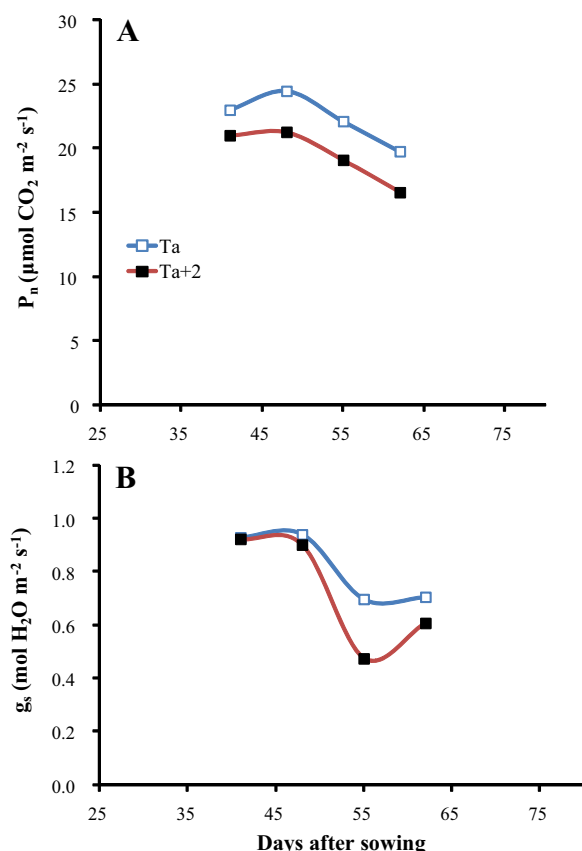


Fig. 4. The response of photosynthesis (A) and stomatal conductance (B) to increased temperature in 2011.

R1 was not affected by increased temperature, except for in 2010 when it was significantly delayed. The temperature during the preflowering period might have largely exceeded the optimum for developmental progress in 2010, the warmest year (data not shown). The R5 stage was consistently delayed by increased temperature in each year. This delay in plant development is a trend different from that shown in previous studies (Craufurd and Wheeler, 2009), which reported enhanced plant development with increased temperatures, including for soybeans (Prasad et al., 2005); however, using a wide range of temperatures, Thomas et al. (2010) found a longer seed-filling period under increased

temperatures when they studied the effect of temperature on seed developmental ontogeny, with 32/22 °C appearing to be a critical optimum. Therefore, in our study, temperatures above optimum appear to have occurred in 2010 and delayed plant development.

Seed yield was reduced under increased temperatures by 16% to 40% in the current study. The reduction resulting from high temperatures was caused by the reduction of important yield components such as fertile pods, seed number, and seed size, and the poor pod-set is presumably related to delayed onset of seed growth.

Decreased soybean seed yields resulting from temperatures >30 °C in controlled-environment chambers have been reported, and were associated with either decreased seed number per plant or decreased single-seed weight (Gibson and Mullen, 1996; Heinemann et al., 2006); however, biomass production was not considerably affected, even at mean temperatures near 40 °C (Allen and Boote, 2000) and the seed yield was not reduced under high-temperature stress induced only at the onset of R5 (Ferris et al., 1999). In contrast, seed yield was reduced under a moderately high temperature of 27.5 °C during the entire growing season (Heinemann et al., 2006). In TGC, Ohe et al. (2007) reported a reduction in yield of approximately 30% as a result of a temperature increase of approximately 4 °C (27–31 °C); Shiraiwa et al. (2006) also found a reduction in yield. In these TGC studies, the detailed mechanisms of reduction were not elucidated. Compared to the above studies, the results obtained in 2009 and 2010 might be unique, because a temperature increase of 3 °C resulted in a yield reduction as great as 40% in 2009 and 38% in 2010.

Reviewing these results, one might suspect that plants under high temperatures encountered transient decreases in water content as an artifact of soil moisture control within the first 2 years. This is because irrigation was conducted in only limited times and occurred at the same time in all the plots, boosting soil moisture to field capacity. Plots were then left to lose soil moisture naturally until the next irrigation to resume field capacity; however, in 2011 and 2012, we managed to ensure that the soil was continuously wet enough to prevent any confounding effects of dehydration. The results of the 2011/2012 study showed similar trends in DM accumulation, pod/seed development, HI, and consequent reduction in seed yield resulting from increased temperature.

One possibility, other than drought occurrence, for this large negative effect of high temperature is the reduction in ambient [CO_2] under high temperature and a low light environment in the TGC. Because of the nature of TGCs, where air flow was created from the entrance (near Ta) to Ta+2 or Ta+3 along the rows of growing plants, there was a reduction in ambient [CO_2], despite the relatively high air flow rate inside the chamber. Thus, the substantial yield reduction with an increase in temperature might be enhanced because these plants were potentially exposed to lower-than-ambient [CO_2] and a 20% reduction in incident radiation through the film in the tunnel. However, the difference in [CO_2] of 20 $\mu\text{L/L}$ mentioned earlier does not appear to contribute in major part to the observed change in photosynthetic rate and DM accumulation.

In addition, high VPD resulting from increased temperatures might have contributed to reduced photosynthesis through stomatal closure part of the time (Table 2 and Fig. 6). There were days between anthesis and R5 when maximum temperatures were as high as 35 °C in 2009 and 36 °C in 2010, and there existed a concomitant increase in VPD up to 2.5 or even 3.0 kPa (Fig. 6). As has been demonstrated (Bunce, 1984; Fletcher et al., 2007), the increase in VPD might result in stomatal closure to a greater extent in the warmer treatment than in the cooler treatment. Lower stomatal conductance was observed in plants grown under high temperature (Fig. 4B). Although it has not been examined for soybean cultivar Enrei, some commercial cultivars tend to restrict transpiration with increasing VPD > 2.1 kPa (Fletcher et al., 2007). Bunce (1984)

Table 3

The response of carbon isotope discrimination (CID) to increased temperature in 2011 and 2012.

Year and treatment	CID	
	Seed (‰)	Leaf (‰)
2011		
Ta	20.39	21.77
Ta + 2	19.50	20.84
2012		
Ta	20.39	21.52
Ta + 2	20.06	21.42
ANOVA	F value (P)	
Treatment (T)	12.09 (<0.05)	5.71 (<0.1)
Year (Y)	3.04 (n.s.)	4.27 (n.s.)
T × Y	3.54 (n.s.)	4.63 (<0.1)

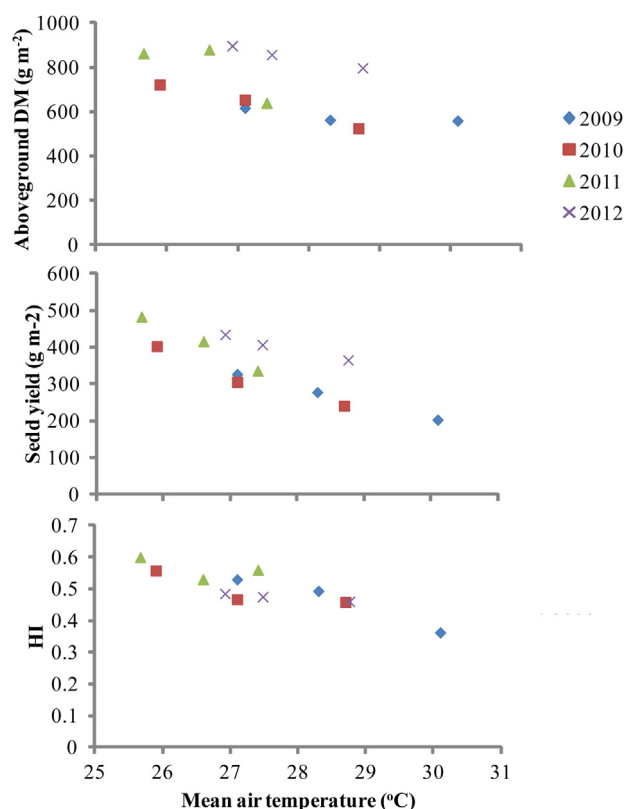


Fig. 5. Total dry matter (DM), seed yield, and harvest index (HI) as affected by temperature.

found that stomatal conductance at a VPD of 3 kPa was lower than that at 1 kPa, and a 50% reduction in stomatal conductance resulted in a 20% reduction in photosynthesis. In our study also, high VPD in high-temperature plots might have affected the photosynthetic rate and yield formation process.

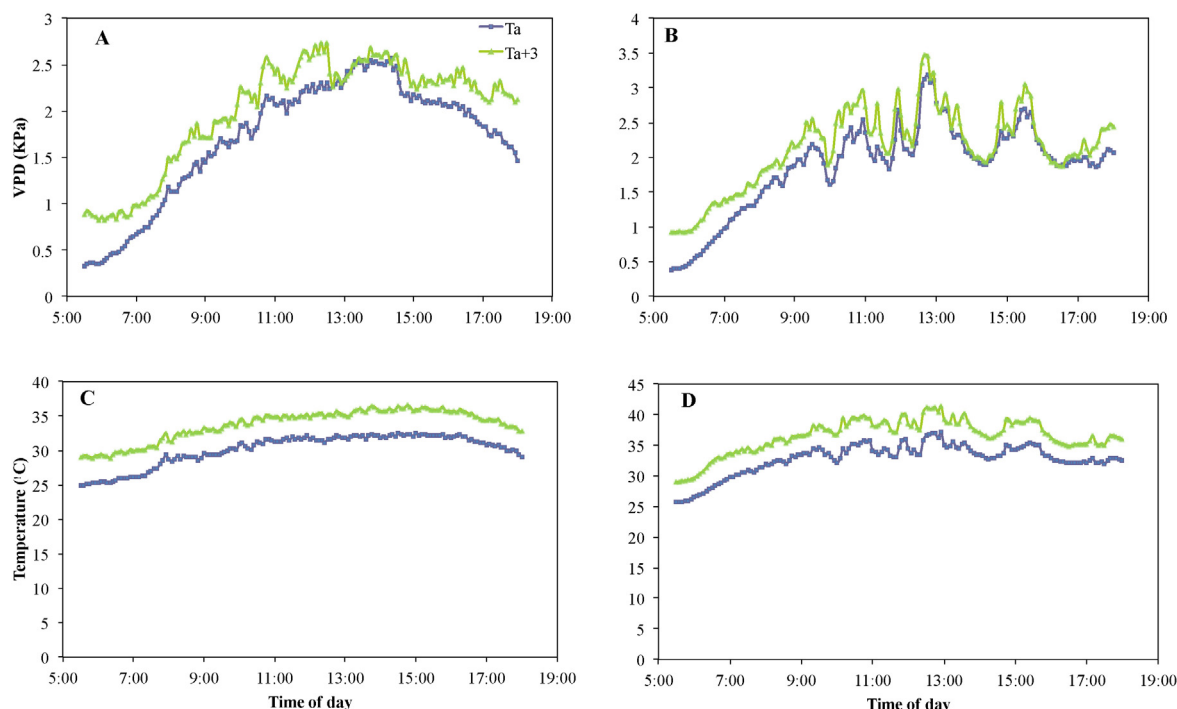


Fig. 6. Vapor pressure deficit (VPD) and temperature on August 14, 2009 (A and C, respectively), and August 17, 2010 (B and D, respectively).

In general, a reduction in P_n is caused not only by reduced g_s but also by reduced mesophyll activity. Considering the coordinating behavior of g_s and mesophyll activity in leaf photosynthesis (Wong et al., 1979), it is difficult to point out g_s as a cause of varied P_n , simply based on the association of the two variables; however, the analysis of CID, a measure of balance between CO_2 fixation in the mesophyll and CO_2 diffusion from the atmosphere into intercellular spaces (Farquhar et al., 1982), showed a significant reduction after high-temperature treatment (Table 3). A reduction in CID with increased temperature partially meets with above-mentioned decline in g_s in the high-temperature treatment, and suggests that reduction in g_s , rather than mesophyll activity, was most likely more responsible for the reduction in DM productivity.

Temporal changes in aboveground DM (Fig. 2) showed that DM accumulation was more affected by the treatments from R1 to R5 than in other periods. Considering that temperatures were highest from R1 to R5 during the growing season (Fig. 1), this also shows that DM accumulation was limited under high temperatures, which might be associated with restricted stomatal conductance.

The effects of temperature might have been aggravated by the fact that high temperatures occurred from flowering to early seed filling. It is generally recognized that the potential impacts of climate change and its resultant temperature change on crop production are particularly evident when high temperatures are induced during the postanthesis phase in soybeans (Hatfield et al., 2011). In the common bean, another crop legume, heat stress from near anthesis to R8 was found to affect the pollen viability and yield of sensitive genotypes (Porch and Jahn, 2001).

In terms of HI, this study confirmed previous findings that temperatures increased to above the optimum decreases DM partitioning to seed in many crops, including soybeans (Boote et al., 2005; Shiraiwa et al., 2006). Indeed, combined data from the four study years show that HI, seed yield, and total DM are consistently reduced by increased temperatures (Fig. 5).

The reduction in HI was associated with a reduced number of pods/seeds and smaller seeds. Because seed size reduction is considered to be the result of reduced cell number, which is determined by mid-seed filling (R5–R6) (Tacarindua et al., 2012), these

processes related to HI reduction are assumed to be a consequence of internal and external factors from after R1 until R5. However, whether the poor pod set and seed development is a result of reduced DM production or a direct effect of temperature remains to be elucidated.

A particular feature of these TGC experiments is that high temperatures caused delayed leaf senescence (or green stem disorder), and this phenomenon was more marked in 2010 than in 2009 (Table 1). In 2010, more green leaves were harvested at maturity and might have compensated the final aboveground DM as compared to 2009, when leaves were old and many had abscised by harvest time.

Considering that soybeans induce senescence primarily to fill their seedpods (Sinclair and de Wit, 1976; Egli, 2004), the delayed pod and seed growth and smaller seeds in the higher temperature treatment might have caused delayed senescence resulting from sink limitation. A temperature of a few degrees higher also caused delayed senescence in some soybean cultivars, and the seeds were smaller in an experiment conducted in the same TGC; however, the magnitude of the delayed senescence resulting from increased temperatures was cultivar dependent, with some cultivars not showing any delayed senescence (Mochizuki et al., 2005). This fact and the results of this study suggest a need to explore genetic variation in the responses of soybean performance to increased temperature.

5. Conclusion

Large effects of an increase in temperature by a few degrees were observed in our TGC studies conducted under field-like conditions at Kyoto, a warm region of Japan. The yield was reduced in plants treated with an increase in temperature through reductions of pod and seed numbers associated with a decline in DM production. It was suggested that the reduction in ambient [CO₂] and light intensity and the concomitant increase of VPD with increased temperature in the TGC exacerbated the effects of increased temperature. The temperature-related reduction in yield is likely to occur in warm regions, where high temperatures coincide with atmosphere dryness and it might occur more often with the climate change; however, whether the dryness of the atmosphere exacerbates the effects of high temperature through drought is a matter for further studies.

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